

ENTRY 064 — Brisbane Dataset to 1,621 — Scale Helps One Direction, Not the Other — September 3, 2026 — Status: COMPLETE

Entry 064 — Does More Brisbane Data Close the Remaining Generalization Gap?

Bell-Pair Dataset Grown 740 → 1,621; Infrastructure Bug Fixed Along the Way

What Was Done

Following Entry 062's finding that a distributional confound only partly explained Brisbane's weak cross-chip transfer, Brisbane's bell-pair dataset was grown from 740 to 1,621 circuits, more than doubling it, to test whether the remaining gap shrinks with scale the way the original two-chip generalization work (Entries 053-058) improved with more data. The 881 new circuits were folded into the graph dataset (4,005 → 4,886), and both cross-chip directions were re-trained and re-evaluated.

A Real Data-Loss Incident, Caught and Fixed

Partway through this growth, a checkpoint write was killed mid-write by the sandbox's hard timeout and corrupted the dataset file, destroying progress back to 66 records (from 900). Root cause: the generator wrote checkpoints via `json.dump` directly to the live file, which truncates the file immediately on open and streams content incrementally -- a kill mid-write leaves a corrupt fragment and destroys the prior good save too, not just the in-progress one. This was caught immediately (not discovered later), the actual damage was assessed honestly (the graph dataset already had 764 circuits permanently baked in from before, so only ~160 records of unconsumed growth were actually lost), and the generator was fixed to write to a temp file and rename atomically before continuing. This is the same class of fix Entries 058/060 applied to circuit generation speed -- here applied to generation safety.

Results

	Entry 062 (764 Brisbane)	Entry 064 (1,645 Brisbane)
Brisbane	MAE 5.23 / R ² =0.354 / fc 25.11 (n_fc=14)	MAE 5.89 / R ² =0.275 / fc 15.57 (n_fc=35)
Sherbrooke	MAE 4.62 / R ² =0.718 / fc 22.85	MAE 3.76 / R ² =0.811 / fc 19.43

Reading This Honestly

One direction improved cleanly with scale; the other did not, and both readings are reported rather than only the favorable one. Brisbane→Kyiv+Sherbrooke improved on every metric and reached its best result across all three entries (R²=0.811, the closest any Brisbane-involving direction has come to the original Kyiv↔Sherbrooke baseline of 0.84–0.88) -- consistent with the hypothesis that more training diversity helps the model learn chip-invariant structure. Kyiv+Sherbrooke→Brisbane did not follow the same pattern: R² moved from 0.354 down to 0.275, even though floor-collapse error on that same test actually improved substantially (25.11 → 15.57) and the floor-collapse test population nearly tripled (14 → 35 examples, a much sounder sample). The most likely explanation is not that the model regressed, but that Brisbane's growth deliberately targeted underfilled long-hop-distance bins (Entry 064's own growth run stalled repeatedly on exactly these, taking dedicated debugging to push through), so the held-out Brisbane test set is now structurally harder -- more long-route, higher-decoherence circuits -- than it was in Entry 062. A test set that got harder because it got more representative of the full distribution is not the same thing as a model that got worse, and the improved floor-collapse number supports that reading. But this is an inference, not a proof, and is recorded as an open question rather than a settled one.

Entry 064 — Conclusion

Brisbane's dataset more than doubled (740 → 1,621 bell pairs, 4,886 graphs combined), with a checkpoint-corruption bug caught and fixed with atomic writes along the way. Scale measurably helped the easier cross-chip direction to its best result yet, and produced a genuinely mixed, honestly-reported result on the harder direction -- worse headline R²

but better floor-collapse accuracy on a much larger, harder test sample. The third-chip generalization question remains open in the same way it has since Entry 061: real progress, not yet resolved.

Research Log — Index Addendum (Entry 064)

Files: entry061_grow_brisbane.py (atomic-write fix), entry063_build_graphs.py, entry063_train_gnn.py, entry063_cross_chip.py.

Data: quantumbridge_data/entry061_brisbane_bell_dataset.json (1,621), entry063_graph_dataset.json (4,886 graphs), entry063_gnn_results.json, entry063_cross_chip_results.json.